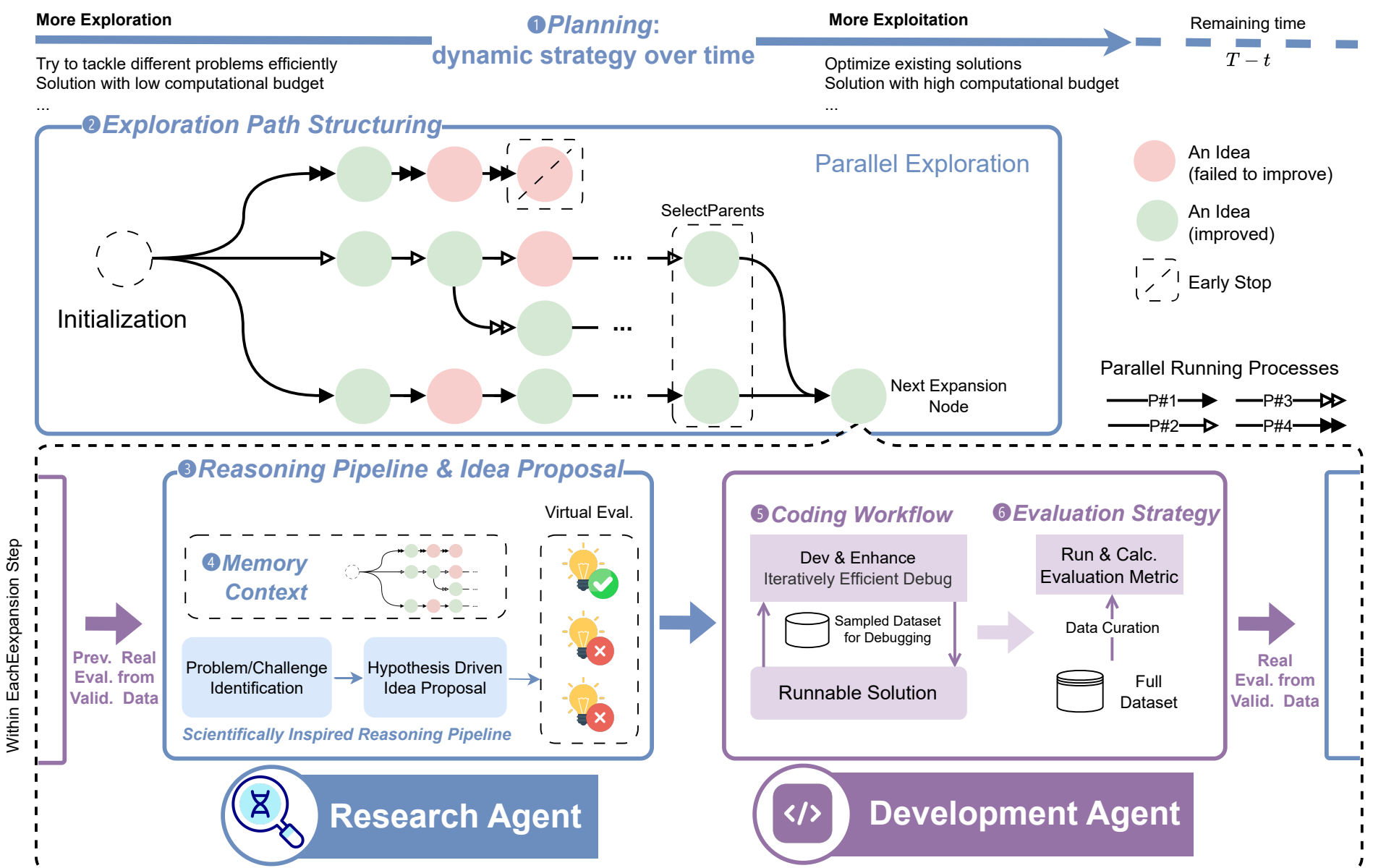
More Exploration
Try to tackle different problems efficiently
Solution with low computational budget
...
❶Planning: dynamic strategy over time
More Exploitation
Optimize existing solutions
Solution with high computational budget
...
Remaining time
$T - t$
❷Exploration Path Structuring
Parallel Exploration
Initialization
SelectParents
Next Expansion Node
An Idea (failed to improve)
An Idea (improved)
Early Stop
Parallel Running Processes
P#1
P#2
P#3
P#4
Within EachEexpansion Step
Prev. Real Eval. from Valid. Data
❸Reasoning Pipeline & Idea Proposal
❹Memory Context
Virtual Eval.
Problem/Challenge Identification
Hypothesis Driven Idea Proposal
Scientifically Inspired Reasoning Pipeline
❺Coding Workflow
Dev & Enhance Iteratively Efficient Debug
Sampled Dataset for Debugging
Runnable Solution
❻Evaluation Strategy
Run & Calc. Evaluation Metric
Data Curation
Full Dataset
Real Eval. from Valid. Data
Research Agent
Development Agent